

MIST: Coordinate-Invariant Spatial Modeling for Gene Expression Prediction from Histology

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Abstract

Predicting spatial gene expression from routine histology images could extend molecular analysis to tissue samples without spatial transcriptomic measurements. This task requires integrating local morphology with tissue context while avoiding sensitivity to arbitrary choices of coordinate origin and orientation. We introduce MIST, a compact spatial transformer that combines three sources of context in each layer: attention over nearby spots, attention over all spots on a slide, and a pooled slide representation. The local pathway uses continuous pairwise distances to bias attention, retaining information about spatial proximity beyond a binary neighborhood graph. Because coordinates enter the model only through these distances, predictions are invariant to translations, rotations, and reflections of the coordinate frame when image features are fixed. We evaluate MIST on seven human organ cohorts from HEST-1k through intra-cohort prediction, pooled multi-organ testing on held-out samples, and leave-one-organ-out cross-organ transfer, using matched features, gene panels, splits, and model-selection procedures within each comparison. In intra-cohort prediction of 100 genes, MIST improves mean Pearson correlation by 10.0% over the strongest baseline and leads on all seven cohorts. In pooled evaluation on held-out samples from represented organs, it improves correlation by 0.3% over the strongest baseline. In a separate 50-gene cross-organ evaluation with an entire organ held out during training, MIST improves correlation by 3.6% over BLEEP, the strongest baseline considered. Across frozen UNI, CONCH, and combined UNI+CONCH features, MIST consistently achieves the highest mean correlation among the methods in the pooled encoder comparison. Factorial ablations identify local attention as the main contributor to predictive accuracy. With 4.88M parameters and model-only inference of 64.7ms for 10,000 spots on an A100, MIST supports accurate, efficient prediction across cohorts and feature encoders, while generalization to unseen organs remains a challenge.

1 Introduction

Spatial transcriptomics (ST) measures gene expression together with tissue location, enabling the study of spatial heterogeneity, tissue organization, and cellular interactions [Ståhl et al. \(2016\)](#). Its cost and experimental burden, however, constrain the number and diversity of specimens that can be profiled. In contrast, H&E slides are routinely collected and contain morphological signals associated with tissue composition and molecular state. Predicting spatial gene expression from these images could therefore extend molecular analysis to specimens without matched ST measurements.

The central difficulty is that morphology and expression are related at more than one level. A spot contains local visual evidence, nearby spots define its tissue neighborhood, distant regions can share structure, and the composition of the slide can shift the expected expression profile. Existing methods capture parts of this context through independent patch regression [He et al. \(2020\)](#), spatial graphs and transformers [Zeng et al. \(2022\)](#), cross-modal retrieval [Xie et al. \(2023\)](#), or conditional generation [Zhu et al. \(2025\)](#). Yet adding spatial context

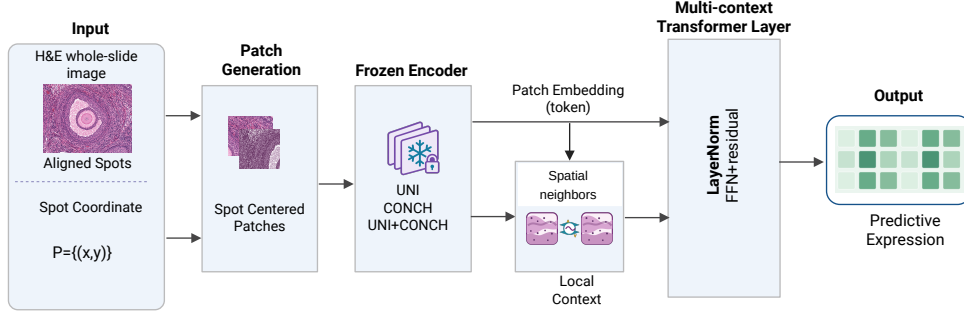
a. MIST Architecture Overview**b. Multi-context Transformer Layer**

Figure 1: MIST overview. (a) From an H&E whole-slide image with aligned spot coordinates $P = \{(x, y)\}$, spot-centered patches are embedded by a frozen pathology encoder (UNI, CONCH, or UNI + CONCH) into per-spot tokens; a spatial k NN over spot coordinates supplies local context. A stack of multi-context transformer layers then predicts the spatial expression map in a single pass. (b) Each layer fuses a local-context pathway (distance-biased attention over k nearest spatial neighbours) and a global-context pathway (all-spot self-attention) by residual addition, followed by LayerNorm/FFN and a linear expression head. Coordinates enter only through pairwise distances, so the spatial pathway is invariant to translations, rotations, and reflections.

introduces a second concern: a model that consumes absolute positions can depend on the scanner coordinate frame. The same relative tissue arrangement may then produce different predictions after its coordinates are translated, rotated, or reflected.

We ask whether a compact predictor can combine neighborhood context and context across the slide without becoming sensitive to an arbitrary coordinate frame. We address this question by reconsidering two common design choices.

Coordinate representation. Spatial models broadly fall into three groups. Methods such as Hist2ST [Zeng et al. \(2022\)](#) embed absolute x, y integer positions, making predictions sensitive to the scanner coordinate frame (rotation drops mean PCC from 1.00 to ≈ 0.5 in our experiments). Other spatial models instead build a binary k NN adjacency matrix from spot coordinates and aggregate neighbor features with uniform weights; because pairwise distances are rotation-invariant, these models are coordinate-frame invariant, but the adjacency discards the actual inter-spot distances, so all neighbors within the graph are treated as equally close regardless of how near or far they actually are. MIST takes a third path: it encodes continuous pairwise distances as radial-basis features that modulate per-head attention weights, preserving full distance information while remaining invariant to coordinate-frame changes by construction.

Context representation. Local aggregation, interactions across the slide, and a summary of slide composition play distinct roles. Representing them as separate pathways lets a model combine these signals while making their individual contributions directly testable through factorial ablation.

Table 1: Architectural components across representative H&E→ST predictors. MIST is the only model that combines local neighborhood attention, slide-wide global attention, a slide-level summary token, coordinate invariance, and continuous (distance-expressive) spatial encoding rather than binary adjacency. • = present, ○ = absent.

Method	Local context	Global context	Slide token	Coord. invariant	Cont. distance
ST-Net He et al. (2020)	○	○	○	○	○
HisToGene Pang et al. (2021)	○	•	○	○	○
Hist2ST Zeng et al. (2022)	•	•	○	○	○
BLEEP Xie et al. (2023)	○	○	○	○	○
TRIPLEX Chung et al. (2024)	•	•	○	○	○
M2OST Wang et al. (2025)	•	○	○	○	○
STEM Zhu et al. (2025)	○	•	○	○	○
MIST (ours)	•	•	•	•	•

What We Do. We represent every measured spot as a token and let it learn from three sources in parallel: nearby spots, all spots on the slide, and a summary of the slide. Coordinates are used only to measure distances between spots. Consequently, changing the origin or orientation of the coordinate system does not change the prediction when the image features remain fixed.

Building on these observations, we introduce MIST, a compact spatial transformer for direct gene expression prediction (Fig. 1). Each layer combines three pathways: k NN attention biased by pairwise distance aggregates local neighborhoods, full self-attention models dependencies across the slide, and attention pooling constructs a slide representation that is broadcast to every spot. The three outputs are fused residually before the expression head. For fixed histology features, MIST is invariant to translations, rotations, and reflections of the coordinate frame because coordinates affect the network only through pairwise distances. This guarantee concerns transformations of the coordinate representation; it does not assume that the frozen image encoder is invariant to transformations of the histology image itself.

We evaluate MIST on seven human organ cohorts from HEST-1k using matched image features, gene panels, data splits, optimization budgets, and nested validation. We distinguish prediction within individual cohorts, pooled training across represented organs, and leave one organ out (LOOO) transfer to an anatomy absent from training. Within cohorts, MIST achieves a mean Pearson correlation of 0.470, compared with 0.426 for the strongest baseline (BLEEP), and leads on all seven cohorts. It reaches 0.760 in pooled evaluation and 0.490 under LOOO transfer. A factorial ablation on SKCM and LUNG identifies local attention as the dominant contributor, while the complete model achieves the highest mean. MIST uses 4.88M parameters and predicts a slide with 10,000 spots in 64.7 ms on an A100.

We investigate five research questions: RQ1 (Accuracy): Does MIST match or exceed existing predictors for intra-cohort H&E→ST prediction across gene-panel sizes (10/50/100) and selection strategies (HVG, DEG, HMMV)? RQ2 (Transfer): How well does MIST generalize across organs under pooled cross-validation and the stricter leave-one-organ-out (LOOO) setting? RQ3 (What matters): Which design choices drive accuracy: the frozen patch encoder, the local/global/slide-level context streams, or the spatial-neighborhood definition? RQ4 (Structure): Do MIST’s predictions preserve coherent spatial tissue structure usable for downstream analysis? RQ5 (Efficiency): Is MIST computationally efficient for whole-slide prediction?

Contributions. This work makes three contributions:

1. We introduce MIST, a multi-scale spatial transformer that fuses local, slide-wide, and slide-summary context in every layer using continuous pairwise-distance RBF features as

its sole coordinate signal, providing coordinate-frame invariance while retaining distance information.

2. We establish the value of this design through rotation-sensitivity tests, factorial context ablations, and distance-encoding controls, identifying the components that drive accuracy.
3. To the best of our knowledge, we provide the first controlled pooled cross-organ evaluation for histology-to-spatial-expression prediction, alongside intra-cohort and held-out-organ tests, with matched features, gene panels, splits, baselines, and model-selection procedures.

2 Related Work

Machine Learning Prediction of Gene Expression from Histology. ST-Net [He et al. \(2020\)](#) established patch-level regression. Hist2ST [Zeng et al. \(2022\)](#) adds spatial context through a graph over neighboring spots, while BLEEP [Xie et al. \(2023\)](#) aligns image and expression embeddings for cross-modal retrieval. STEM [Zhu et al. \(2025\)](#) instead treats expression prediction as conditional generation. These methods use different features, gene panels, splits, and selection procedures, so their published scores do not isolate architectural gains; our comparisons therefore use a matched protocol. MIST is a deterministic point predictor and does not model the full conditional expression distribution.

Geometric Structure and Coordinate Invariance. Spatial prediction depends on the coordinate representation supplied to a model. Absolute position embeddings can encode location within a particular coordinate frame, but their output may change when an equivalent tissue layout is translated, rotated, or reflected. Geometric networks address this issue by constructing interactions from relative quantities [Satorras et al. \(2021\)](#). MIST uses pairwise Euclidean distances, which are preserved by translations, rotations, and reflections. The guarantee applies to the spatial pathway for fixed histology features, not to rotations of the image encoder input.

Generalization across Samples and Organs. HEST-1k [Jaume et al. \(2024\)](#) spans many cohorts and organs. Within-cohort evaluation can exploit organ-specific morphology or acquisition effects. We therefore distinguish pooled cross-validation, which holds out samples while retaining their organ types in training, from LOOO, which withholds an entire organ and provides a stricter test of anatomical transfer.

Positioning of MIST. MIST connects these three research directions: it is an inductive-bias-explicit spatial transformer for direct regression, evaluated under a matched cross-cohort protocol. Unlike absolute position encodings or frame-averaging strategies, its coordinate pathway is invariant by construction. Unlike generative models, it does not represent expression uncertainty; its advantage is deterministic, single-pass inference with explicit geometric structure. Unlike prior evaluations that report within-cohort accuracy only, our controlled comparison separates interpolation from anatomical transfer, making the difficulty of organ shift directly measurable.

3 Method

MIST maps frozen spot-level histology features and measured spot coordinates to a spatial gene expression map. Its method has three stages: coordinate-invariant distance encoding, a shared multi-context layer, and supervised prediction with a correlation-aware objective.

3.1 Problem Setup

For a slide with N spots, let $\mathbf{F} = [\mathbf{f}_1, \dots, \mathbf{f}_N]^\top \in \mathbb{R}^{N \times d_{\text{in}}}$ denote frozen pathology features and $\mathbf{P} = [\mathbf{p}_1, \dots, \mathbf{p}_N]^\top \in \mathbb{R}^{N \times 2}$ the corresponding spot coordinates. The target $\mathbf{Y} \in \mathbb{R}^{N \times G}$ contains log-normalized expression for G genes. We learn $f_\theta : (\mathbf{F}, \mathbf{P}) \mapsto \hat{\mathbf{Y}}$ and process each

slide independently; the model therefore supports a variable number of spots and does not require expression measurements at test time. In the feature-matched main comparisons, $\mathbf{f}_i$ concatenates frozen UNI Chen et al. (2024) and CONCH Lu et al. (2024) embeddings ($d_{\text{in}} = 1536$).

3.2 Coordinate-Invariant Distance Encoding

We first compute the pairwise Euclidean distance matrix $D_{ij} = \|\mathbf{p}_i - \mathbf{p}_j\|_2$. For each spot i , we retain the indices $\mathcal{N}(i)$ and distances of its k nearest neighbors, excluding the spot itself. Distances are expanded into M radial-basis features,

$$\phi(D_{ij}) = [\exp(-\gamma_m D_{ij}^2)]_{m=1}^M, \quad (1)$$

where the fixed bandwidth parameters γ_m are logarithmically spaced. These features influence attention weights but are never added to the spot embeddings as absolute positions.

Proposition 1 (Coordinate frame invariance):

Let $R \in \mathbb{R}^{2 \times 2}$ be orthogonal, let $\mathbf{t} \in \mathbb{R}^2$, and define $\mathbf{P}' = \mathbf{P}R^\top + \mathbf{1}\mathbf{t}^\top$. For fixed histology features $\mathbf{F}$, the prediction is unchanged:

$$f_\theta(\mathbf{F}, \mathbf{P}') = f_\theta(\mathbf{F}, \mathbf{P}). \quad (2)$$

Thus, the spatial pathway is invariant to translations, rotations, and reflections of the coordinate frame in exact arithmetic. The guarantee concerns a change of coordinate frame with fixed histology features; it does not claim that the frozen image encoder is invariant to transformations of the histology image. The proof is in Appendix I. Note that binary k NN adjacency models satisfy the same invariance property, since pairwise distances are also rotation-invariant. The distinguishing property of MIST is that it retains the magnitude of those distances as continuous RBF features influencing attention weights, whereas binary-adjacency models collapse each distance to a 0/1 neighbor indicator, losing the information of how close or far each neighbor actually is.

3.3 Multi-Context MIST Layer

Core Idea. At every layer, each spot receives three messages in parallel: one from its nearby spots, one from all spots, and one from a pooled summary of the slide. The model adds these messages to the current spot representation and refines the result with a feed-forward network. Only the nearby-spot message uses coordinates, and it uses pairwise distances rather than absolute positions.

The input features are projected to hidden width d , $\mathbf{X}^{(0)} = \mathbf{F}W_{\text{in}} + \mathbf{b}_{\text{in}}$. The same neighbor graph and radial-basis features are reused in all L transformer layers. At layer ℓ , a pre-normalized representation $\mathbf{H} = \text{LN}(\mathbf{X}^{(\ell)})$ is passed through three context pathways.

3.3.1 Local distance-biased attention

Each spot attends to its k neighbors with a learned, distance-dependent bias for each attention head. For head h , the attention weights are

$$\alpha_{ij}^h = \text{softmax}_{j \in \mathcal{N}(i)} \left(\frac{(\mathbf{q}_i^h)^\top \mathbf{k}_j^h}{\sqrt{d_h}} + b_h(\phi(D_{ij})) \right), \quad (3)$$

where $d_h = d/H$ for H heads. The local output is the concatenation of the weighted value sums, followed by a learned output projection. Morphological similarity is captured by the query-key term, while spatial distance contributes through the learned radial-basis bias.

3.3.2 Global self-attention

In parallel, standard multi-head self-attention [Vaswani et al. \(2017\)](#) allows every spot to attend to all N spots. This pathway captures long-range dependencies that may not be connected in the local graph. It uses morphology features only and is permutation-equivariant with respect to spot order.

3.3.3 Attention-pooled slide context

A learned scalar scoring function produces $a_i = \text{softmax}_i(s(\mathbf{h}_i))$. The slide representation is

$$\mathbf{g} = W_g \left(\sum_{i=1}^N a_i \mathbf{h}_i \right) + \mathbf{b}_g, \quad (4)$$

and is broadcast to every spot. Unlike a learned class token, this representation is computed directly from the slide’s spot features and summarizes its morphological composition.

For the complete V3 model, the local output, global output, and broadcast slide representation are summed and added to the residual stream. A second pre-normalization residual update applies a two-layer feed-forward network with expansion factor four and GELU activation:

$$\tilde{\mathbf{X}}^{(\ell)} = \mathbf{X}^{(\ell)} + \mathbf{L}^{(\ell)} + \mathbf{A}^{(\ell)} + \mathbf{G}^{(\ell)}, \quad (5)$$

$$\mathbf{X}^{(\ell+1)} = \tilde{\mathbf{X}}^{(\ell)} + \text{FFN}(\text{LN}(\tilde{\mathbf{X}}^{(\ell)})). \quad (6)$$

After L layers, layer normalization and a linear head map each spot representation to its G predicted gene values. The model produces the complete slide prediction in one forward pass.

Proposition 2 (Permutation equivariance):

Let $Q \in \{0, 1\}^{N \times N}$ be a permutation matrix. If the k NN construction handles distance ties equivariantly, then

$$f_\theta(Q\mathbf{F}, Q\mathbf{P}) = Qf_\theta(\mathbf{F}, \mathbf{P}). \quad (7)$$

Therefore, reordering the input spots only reorders their predicted expression profiles. Appendix I provides the proof. Our numerical checks evaluate both Propositions 1 and 2 under finite precision.

3.4 Training Objective and Ablations

For a slide with measured expression $\mathbf{Y}$, we optimize mean squared error together with a gene-wise Pearson-correlation term:

$$\mathcal{L} = \text{MSE}(\hat{\mathbf{Y}}, \mathbf{Y}) + \lambda \left(1 - \frac{1}{|\mathcal{V}|} \sum_{g \in \mathcal{V}} \rho_g \right), \quad (8)$$

where ρ_g is the Pearson correlation between predicted and measured expression across the N spots, $\mathcal{V}$ is the set of genes with nonzero target variance on that slide, and $\lambda = 0.5$. MSE constrains the scale of the predictions, while the correlation term encourages the model to reproduce spot-to-spot expression patterns.

The main model enables all three context pathways. Our factorial ablation independently toggles the local (L), global (G), and slide-level (S) pathways, avoiding conclusions tied to a particular order of architectural additions (Table 7).

4 Experimental Results

Dataset and Preprocessing. We evaluate on HEST-1k [Jaume et al. \(2024\)](#). The controlled human-tissue benchmark retains seven cohorts representing seven organs; HCC,

Table 2: Intra-cohort gene expression prediction on HEST-1k across gene-selection strategies (Pearson $\uparrow$). HVG: highly variable genes (the HEST default); DEG: differentially expressed genes (Leiden clustering + Wilcoxon rank-sum); HMHVG: high-mean highly variable genes selected from variance-filtered candidates using expression statistics. Fifty genes each; mean_{std} over three seeds with nested-validation; best per cohort and strategy in bold. Same features, splits, and training budget as Appendix Table 11. Error metrics are reported separately in Table 3.

Panel	Cohort	ST-Net NBE'20	Hist2ST BIB'22	BLEEP NeurIPS'23	STEM ICLR'25	MIST (Ours)
HVG	IDC Breast	0.551 $\pm$ 0.01	0.305 $\pm$ 0.11	0.519 $\pm$ 0.00	0.434 $\pm$ 0.00	0.581$\pm$0.01
	COAD Colorectal	0.235 $\pm$ 0.01	0.298 $\pm$ 0.01	0.322$\pm$0.01	0.177 $\pm$ 0.01	0.318 $\pm$ 0.01
	CCRCC Kidney	0.163 $\pm$ 0.02	0.180 $\pm$ 0.03	0.175 $\pm$ 0.02	0.118 $\pm$ 0.01	0.252$\pm$0.01
	LUNG Lung	0.521 $\pm$ 0.01	0.430 $\pm$ 0.05	0.565 $\pm$ 0.01	0.407 $\pm$ 0.00	0.589$\pm$0.01
	PAAD Pancreas	0.366 $\pm$ 0.03	0.248 $\pm$ 0.04	0.412 $\pm$ 0.03	0.233 $\pm$ 0.02	0.437$\pm$0.03
	PRAD Prostate	0.287 $\pm$ 0.02	0.323 $\pm$ 0.01	0.346 $\pm$ 0.01	0.207 $\pm$ 0.02	0.410$\pm$0.01
	SKCM Skin	0.661 $\pm$ 0.01	0.523 $\pm$ 0.00	0.650 $\pm$ 0.00	0.461 $\pm$ 0.02	0.707$\pm$0.01
	Average	0.398	0.330	0.427	0.291	0.471
DEG	IDC Breast	0.697 $\pm$ 0.01	0.549 $\pm$ 0.07	0.731 $\pm$ 0.01	0.684 $\pm$ 0.02	0.774$\pm$0.01
	COAD Colorectal	0.497 $\pm$ 0.00	0.608 $\pm$ 0.01	0.639 $\pm$ 0.02	0.534 $\pm$ 0.01	0.677$\pm$0.00
	CCRCC Kidney	0.312 $\pm$ 0.01	0.335 $\pm$ 0.01	0.304 $\pm$ 0.00	0.253 $\pm$ 0.01	0.402$\pm$0.01
	LUNG Lung	0.566 $\pm$ 0.01	0.518 $\pm$ 0.01	0.621 $\pm$ 0.01	0.440 $\pm$ 0.01	0.662$\pm$0.00
	PAAD Pancreas	0.425 $\pm$ 0.03	0.352 $\pm$ 0.01	0.487 $\pm$ 0.03	0.306 $\pm$ 0.04	0.518$\pm$0.03
	PRAD Prostate	0.413 $\pm$ 0.00	0.437 $\pm$ 0.01	0.466 $\pm$ 0.00	0.355 $\pm$ 0.03	0.539$\pm$0.01
	SKCM Skin	0.752 $\pm$ 0.01	0.656 $\pm$ 0.00	0.746 $\pm$ 0.03	0.617 $\pm$ 0.02	0.794$\pm$0.00
	Average	0.523	0.494	0.571	0.456	0.624
HMHVG	IDC Breast	0.724 $\pm$ 0.01	0.633 $\pm$ 0.08	0.739 $\pm$ 0.02	0.687 $\pm$ 0.01	0.774$\pm$0.01
	COAD Colorectal	0.503 $\pm$ 0.01	0.659 $\pm$ 0.02	0.645 $\pm$ 0.00	0.528 $\pm$ 0.03	0.698$\pm$0.01
	CCRCC Kidney	0.339 $\pm$ 0.01	0.387 $\pm$ 0.01	0.349 $\pm$ 0.01	0.290 $\pm$ 0.01	0.454$\pm$0.02
	LUNG Lung	0.568 $\pm$ 0.00	0.598 $\pm$ 0.00	0.661 $\pm$ 0.00	0.517 $\pm$ 0.01	0.693$\pm$0.00
	PAAD Pancreas	0.441 $\pm$ 0.03	0.341 $\pm$ 0.04	0.493 $\pm$ 0.02	0.305 $\pm$ 0.03	0.524$\pm$0.02
	PRAD Prostate	0.452 $\pm$ 0.00	0.477 $\pm$ 0.01	0.432 $\pm$ 0.04	0.337 $\pm$ 0.02	0.555$\pm$0.02
	SKCM Skin	0.824 $\pm$ 0.03	0.710 $\pm$ 0.14	0.849 $\pm$ 0.02	0.731 $\pm$ 0.06	0.901$\pm$0.00
	Average	0.550	0.544	0.595	0.485	0.657

All strategies select 50 genes per cohort. MIST attains the highest average under every strategy; absolute accuracy rises HVG<DEG<HMHVG for all methods, as HMHVg panels are the most predictable from H&E.

Table 3: Error metrics across gene-selection strategies on HEST-1k. Mean_{std} over three seeds with nested-validation; lower values are better and the best result in each row is in bold. The models, features, splits, and training budget match Table 2.

Panel	Metric	ST-Net NBE'20	Hist2ST BIB'22	BLEEP NeurIPS'23	STEM ICLR'25	MIST (Ours)
HVG	Avg. MSE $\downarrow$	2.041	1.653	1.445	1.874	1.388
DEG	Avg. MSE $\downarrow$	3.853	1.961	1.783	2.172	1.724
HMHVG	Avg. MSE $\downarrow$	4.005	1.977	1.754	2.158	1.622
HVG	Avg. MAE $\downarrow$	0.972	0.907	0.843	0.944	0.829
DEG	Avg. MAE $\downarrow$	1.548	1.071	1.003	1.105	0.989
HMHVG	Avg. MAE $\downarrow$	1.549	1.079	0.987	1.094	0.959

Table 4: Cross-organ LOOO transfer, per-cohort Pearson $\uparrow$ (HEST-1k). Mean_{std} over three seeds; nested-validation. Leave-one-organ-out: trained on all other organ groups, tested on the held-out cohort. All methods use identical frozen UNI + CONCH features (1536-dim) and 50-gene panels. Best per row in bold.

Cohort	ST-Net NBE'20	Hist2ST BIB'22	BLEEP NeurIPS'23	STEM ICLR'25	MIST+CONCH
IDC Breast	0.198 $\pm$ 0.05	0.458 $\pm$ 0.01	0.529$\pm$0.02	0.420 $\pm$ 0.03	0.520 $\pm$ 0.02
COAD Colorectal	0.420 $\pm$ 0.06	0.670 $\pm$ 0.03	0.662 $\pm$ 0.01	0.452 $\pm$ 0.02	0.729$\pm$0.03
CCRCC Kidney	0.068 $\pm$ 0.01	0.055 $\pm$ 0.01	0.095 $\pm$ 0.01	0.090 $\pm$ 0.01	0.109$\pm$0.01
LUNG Lung	0.493 $\pm$ 0.07	0.378 $\pm$ 0.18	0.624 $\pm$ 0.01	0.512 $\pm$ 0.03	0.675$\pm$0.01
PAAD Pancreas	0.459 $\pm$ 0.02	0.399 $\pm$ 0.02	0.537 $\pm$ 0.02	0.409 $\pm$ 0.03	0.548$\pm$0.01
PRAD Prostate	0.079 $\pm$ 0.00	0.098 $\pm$ 0.02	0.118 $\pm$ 0.01	0.046 $\pm$ 0.00	0.118$\pm$0.01
SKCM Skin	0.520 $\pm$ 0.16	0.736 $\pm$ 0.01	0.748$\pm$0.01	0.631 $\pm$ 0.03	0.728 $\pm$ 0.04
Average	0.319	0.399	0.473	0.366	0.490

STEM: corrected DiT ($T = 1000$, 34M params), three seeds.

Table 5: Cross-organ LOOO transfer on a five-organ subset (kidney, breast, lung, prostate, skin; COAD and PAAD excluded), per-cohort Pearson $\uparrow$. Self-contained leave-one-organ-out: each fold trains on the other four organs of the subset, with leakage-safe per-fold gene panels. Identical frozen UNI + CONCH features (1536-dim); mean_{std} over three seeds; best per row in bold.

Cohort	ST-Net NBE'20	Hist2ST BIB'22	BLEEP NeurIPS'23	STEM ICLR'25	MIST+CONCH
CCRCC Kidney	0.099 $\pm$ 0.01	0.095 $\pm$ 0.00	0.061 $\pm$ 0.01	0.032 $\pm$ 0.02	0.120$\pm$0.00
IDC Breast	0.236 $\pm$ 0.03	0.320 $\pm$ 0.01	0.458$\pm$0.00	0.293 $\pm$ 0.00	0.356 $\pm$ 0.07
LUNG Lung	0.488 $\pm$ 0.12	0.547 $\pm$ 0.05	0.706 $\pm$ 0.01	0.380 $\pm$ 0.13	0.722$\pm$0.02
PRAD Prostate	0.056 $\pm$ 0.01	0.047 $\pm$ 0.00	0.052 $\pm$ 0.01	0.020 $\pm$ 0.01	0.075$\pm$0.01
SKCM Skin	0.459 $\pm$ 0.08	0.411 $\pm$ 0.19	0.716 $\pm$ 0.02	0.385 $\pm$ 0.14	0.723$\pm$0.02
Average	0.267	0.284	0.399	0.222	0.399

LYMPH_IDC, and READ are excluded for the data-quality and split constraints documented in Appendix H. Each cohort uses the provided patient-stratified folds. We predict the 50 most highly variable genes (HVGs), which are ranked by cross-spot expression variance, after $\log(1+x)$ normalization, using the same frozen UNI+CONCH features for every method in the main comparison. HVGs are the HEST-1k default panel and capture broad tissue-level expression variation.

Gene selection. Published methods use different target-gene selection protocols. ST-Net selects 250 genes with the highest mean expression (highly expressed genes, HEGs) He et al. (2020). Hist2ST selects the top 1,000 HVGs per section before expression-frequency filtering Zeng et al. (2022). BLEEP trains on a union of per-slide HVGs and reports results on HVG, HEG, and marker-gene subsets Xie et al. (2023). STEM evaluates high-mean highly variable gene (HMHVG) and HVG panels on HEST, and HMHVG and differentially expressed gene (DEG) panels on HER2ST; its DEGs are selected across annotated tissue regions in training slides Zhu et al. (2025). Thus, HMHVG in these papers denotes selection by expression mean and variability.

Our panel comparison uses three 50-gene target sets shared by all evaluated models: HVG, the default HEST panel; DEG, selected by Wilcoxon tests across unsupervised Leiden clusters; and HMHVG, selected using expression mean and variability after variance-based candidate filtering. The models are trained and evaluated separately for each panel with the same cohort splits. Table 2 and Appendix G report the comparison.

Experiment Setup. Evaluation regimes. Intra-cohort evaluation trains a separate model within each cohort. Pooled five-fold cross-validation trains jointly across organs and holds out slide-level samples while retaining each organ type in training. Organ identity is used to

Table 6: Cross-organ POOLED on HEST-1k. 5-fold pooled cross-validation; mean_{std} over three seeds; nested-validation. Best per metric in bold.

Method	PCC $\uparrow$	MSE $\downarrow$	RMSE $\downarrow$	MAE $\downarrow$
ST-Net He et al. (2020) (NBE'20)	0.370 _{0.01}	0.944 _{0.05}	0.965 _{0.02}	0.653 _{0.01}
Hist2ST Zeng et al. (2022) (BIB'22)	0.737 _{0.01}	0.467 _{0.01}	0.676 _{0.01}	0.419 _{0.01}
BLEEP Xie et al. (2023) (NeurIPS'23)	0.758 _{0.01}	0.427_{0.02}	0.648_{0.01}	0.380_{0.00}
STEM Zhu et al. (2025) (ICLR'25)	0.643 _{0.01}	0.594 _{0.01}	0.766 _{0.01}	0.428 _{0.01}
MIST (V3)	0.760_{0.01}	0.440 _{0.02}	0.657 _{0.02}	0.410 _{0.01}

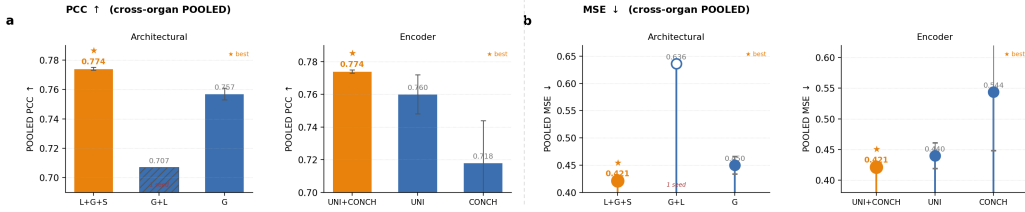


Figure 2: Ablation study on cross-organ POOLED (11-gene leakage-safe panel), grouped by metric: (a) Pearson correlation (PCC $\uparrow$, bars) and (b) mean-squared error (MSE $\downarrow$, lollipops). Each metric is shown for the architectural (context-stream) ablation — global attention only (G), global + local (G+L), and the full model (L+G+S; local + global + slide) — and the feature-encoder ablation — UNI, CONCH, and UNI+CONCH frozen features. The full model / UNI+CONCH is best on both metrics (PCC 0.774, MSE 0.421; starred, $\star$ orange). G+L is drawn hatched/hollow as it currently has a single seed (its remaining stream configurations are still training). Error bars are std over three seeds.

define and report the cross-organ protocol, but organ labels and other slide metadata are not provided as model inputs. LOOO trains on six organs and tests the seventh, which is absent from training. Pooled folds are sample-disjoint; because patient identifiers are unavailable for some cohorts, we do not claim that they are patient-disjoint. To the best of our knowledge, this is the first pooled cross-organ evaluation of histology-to-spatial-expression prediction under a matched, controlled protocol.

Metrics and protocol. We report the Pearson correlation coefficient (PCC), Spearman correlation (SCC), and mean squared error (MSE) between predicted and measured expression, averaged over folds and three random seeds (mean_{std}). Every model selects its checkpoint on an inner-validation split of the training fold and is evaluated once on the outer test-fold; where a cohort fold contains a single training slide, the inner split holds out a disjoint subset of that slide’s spots. This removes the test-fold model selection present in earlier exploratory runs.

Baselines. We compare with ST-Net [He et al. \(2020\)](#), Hist2ST [Zeng et al. \(2022\)](#), BLEEP [Xie et al. \(2023\)](#), and STEM [Zhu et al. \(2025\)](#) where results are available. Features, gene panels, splits, optimization budgets, and checkpoint selection are matched in the controlled tables. Published MCToGene results [Sun et al. \(2026\)](#), which use a different ten-cohort protocol, appear separately in Appendix F and are not treated as directly comparable.

Implementation. MIST uses $L=4$ layers, hidden width $d=256$, 4 attention heads, $k=8$ neighbors, and 16 radial-basis functions (4.8M parameters). We train per slide with AdamW (learning rate 10^{-3} , weight decay 10^{-2}), a cosine schedule, and gradient clipping at 1.0, for at most 100 epochs with early-stopping patience of 20; each step samples at most 3,000 spots. Experiments run on one NVIDIA A100 GPU.

4.1 Results for RQ1: Intra-cohort prediction

Appendix Table 11 reports the feature-matched comparison across gene-panel sizes. For PCC@50, MIST obtains the highest mean PCC (0.470), compared with 0.426 for BLEEP, the strongest baseline, a 10.3% relative improvement. It leads on all seven cohorts, though the margin is not uniform: it is widest on kidney and breast (+0.068 and +0.062 over the best baseline) and narrowest on colorectal (+0.004). The same ranking holds under a smaller 10-gene panel (PCC@10 in Appendix Table 11): with everything else fixed and only the panel reduced to the ten highest-variance genes of each cohort’s HEST 50-HVG set, MIST again leads the average (0.569 vs. 0.526 for BLEEP) and tops five of seven cohorts. Absolute correlations rise across all methods because the retained genes are the most variable, and hence the most predictable.

4.2 Results for RQ2: Cross-organ transfer

LOOO is substantially harder because the target anatomy is absent from training. MIST reaches a mean PCC of 0.490 (Table 4), compared with 0.473 for BLEEP, the strongest baseline, and leads on five of seven held-out cohorts; BLEEP is stronger on IDC and SKCM. Under pooled cross-validation, where slide-level samples are held out while organ types remain represented in training, MIST reaches 0.760 (Table 6), narrowly above BLEEP at 0.758. The gap between pooled (0.760) and LOOO (0.490) directly quantifies the remaining difficulty of unseen-organ anatomical transfer. Appendix A reports a complementary seven-organ mouse experiment.

4.3 Results for RQ3: What drives accuracy and coordinate invariance?

The factorial study in Table 7 evaluates all eight combinations of local (L), global (G), and slide-level (S) context on SKCM and LUNG. Local attention provides the largest gain: enabling it alone raises mean PCC from 0.608 to 0.640. Global and slide context have little effect without the local pathway, but the complete model achieves the highest mean (0.643). This result supports distance-biased neighborhood aggregation as the primary contributor in the tested intra-cohort setting; claims about component importance under organ shift require a matched LOOO ablation not reported here.

Table 8 numerically verifies the analytic invariance result. Differences under translation and rotation remain below 6×10^{-5} and are attributable to floating-point evaluation; reflection is exact in this test.

Table 7: Factorial context ablation (intra-cohort, Pearson mean_{std} over three seeds on SKCM and LUNG). L, G, and S denote local attention, global attention, and slide representation. Local attention is the dominant contributor within an organ.

Variant	L	G	S	SKCM	LUNG	Avg
C000				0.639 _{0.00}	0.577 _{0.01}	0.608
C001			•	0.633 _{0.01}	0.578 _{0.00}	0.605
C010		•		0.641 _{0.01}	0.577 _{0.00}	0.609
C011		•	•	0.639 _{0.02}	0.583 _{0.00}	0.611
C100	•			0.693 _{0.01}	0.587 _{0.00}	0.640
C101	•		•	0.695 _{0.01}	0.585 _{0.01}	0.640
C110	•	•		0.693 _{0.01}	0.584 _{0.00}	0.638
C111	•	•	•	0.696_{0.00}	0.589_{0.01}	0.643

Table 8: Numerical coordinate-invariance check. Maximum absolute prediction difference when only the spot coordinates are transformed (randomly initialized model, 257-spot synthetic slide).

Transformation	Max. abs. difference
Translation	5.97×10^{-5}
Rotation	1.97×10^{-5}
Reflection	0
Spot permutation	5.96×10^{-7}

Intra-cohort spatial gene expression: Skin (SKCM), slide TENX117

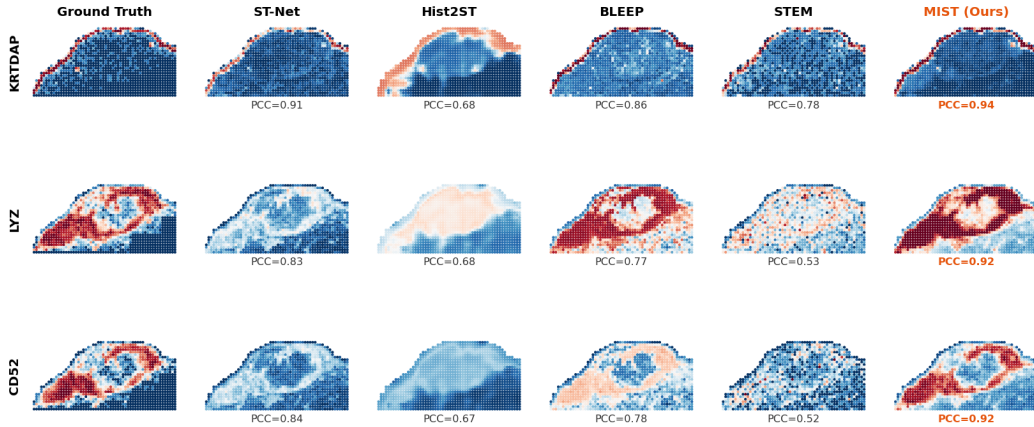


Figure 3: Qualitative spatial expression maps for a held-out skin (SKCM) slide (TENX117), three genes (rows). Each panel colors every spot by expression; MIST (right) most faithfully recovers the ground-truth spatial structure: the perivascular ring for LYZ/CD52 and the epidermal edge for KRTDAP. It attains the highest per-gene PCC (in orange), whereas STEM is noisy and Hist2ST over-smooths. Predictions use identical frozen UNI+CONCH features.

4.4 Results for RQ4: Spatial structure preservation

Gene-wise PCC does not test whether predictions preserve coherent tissue regions. We cluster spots using either predicted or measured expression and compare the induced domains with adjusted Rand index (ARI) and NMI. MIST leads on both ARI (0.184) and NMI (0.239), ahead of the strongest baseline (ST-Net at 0.154 and 0.212; Appendix B). The differences remain small and slide-level variation is large, so we caution against over-interpreting them as evidence of uniformly improved domain recovery.

4.5 Results for RQ5: Computational efficiency

MIST contains 4.88M parameters and predicts a complete slide in one forward pass (Table 9). Its advantage over the baselines is not minimum parameter count: ST-Net and BLEEP are smaller, whereas MIST combines multi-scale spatial context with direct inference. Model-only latency grows from 3.5 ms for 500 spots to 64.7 ms for 10,000 spots on an A100, with 3.1 GiB peak allocated memory at the largest size (Table 10); feature extraction and disk I/O are excluded. STEM’s iterative diffusion inference is the only approach with substantially higher latency.

Table 9: Model complexity. Parameter counts and inference mechanisms.

Model	#Params (M)	Inference
ST-Net He et al. (2020) (NBE’20)	0.47	single-pass
Hist2ST Zeng et al. (2022) (BIB’22)	2.18	single-pass
BLEEP Xie et al. (2023) (NeurIPS’23)	0.54	single-pass
STEM Zhu et al. (2025) (ICLR’25)	34.16	iterative
MIST (V3)	4.88	single-pass

Table 10: MIST inference scaling on an NVIDIA A100 80GB. Median model-only latency and peak allocated GPU memory over synthetic slides; feature extraction and disk I/O are excluded.

Spots	Latency (ms)	Memory (MiB)
500	3.48	43
1,000	3.63	67
3,000	9.36	335
5,000	21.31	842
10,000	64.74	3,183

5 Discussion

Three findings merit emphasis. First, a compact direct predictor is competitive when features, gene panels, splits, and model selection are controlled; the 10.3% relative gain over the strongest baseline rests on matched experimental conditions, not a larger model. Second, pooled multi-organ performance is not a substitute for unseen-organ evaluation: the drop from 0.760 (pooled) to 0.490 (LOOO) shows that anatomical shift remains a separate and harder problem. Third, local distance-biased attention is the dominant contributor in the factorial study; global and slide pathways provide moderate complementary gains, and the complete model (C111) achieves the best mean across both tested cohorts.

Direct and generative models serve different purposes. STEM [Zhu et al. \(2025\)](#) models a conditional expression distribution and can represent one-to-many uncertainty through iterative sampling. MIST produces a point estimate optimized for correlation and squared error. Its benefits, including deterministic, single-pass inference and explicit coordinate invariance, do not imply that morphology uniquely determines expression.

Limitations. The evaluation predicts 50 genes from frozen UNI+CONCH embeddings; results may not extend to transcriptome-scale output or end-to-end image encoding. LOOO uses a fixed benchmark panel and does not test platform or institution shift simultaneously. Coordinate invariance holds only for the spatial pathway with fixed image features; rotating the histology image can change its frozen embedding. Domain-recovery analysis covers only two baselines and has high slide-level variance. Pooled folds are sample-disjoint but cannot be verified as patient-disjoint where patient identifiers are unavailable.

6 Conclusion and Future Work

We developed MIST, a coordinate-invariant local-global spatial transformer for predicting gene expression from histology. Pairwise-distance encoding makes the spatial pathway invariant to rigid coordinate transformations by construction, while three parallel context signals: local k NN attention, global self-attention, and slide-level pooling, together produce complete-slide predictions in a single pass. These design choices directly address our design goals that multi-scale context is necessary and that architectural positional bias should not break under coordinate shift. Under matched HEST-1k evaluation, MIST improves mean intra-cohort correlation and remains competitive in LOOO transfer; the pooled-to-LOOO gap exposes the remaining difficulty of anatomical transfer as the central open challenge.

For future work, it would be valuable to explore how conditioning on slide-level metadata (organ type, sequencing platform) can narrow this anatomical gap, for instance, through FiLM or adaptive layer normalisation conditioners that are available at inference time without requiring expression labels. Scaling the gene panel beyond 50 genes, incorporating end-to-end image encoding, and combining MIST’s geometric structure with calibrated generative uncertainty modeling (as in STEM) are further natural directions.

AI use statement

Generative AI tools assisted with manuscript organization, language editing, ICLR formatting, and consistency checks between narrative claims and reported tables. They were not used to generate experimental data or execute model experiments. The authors reviewed all AI-assisted revisions and take responsibility for the final content of this work.

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A Mouse cross-organ diagnostic

We evaluate 59 mouse Visium slides spanning seven organs with mouse-specific 50-gene panels. Pooled performance reaches 0.887, whereas every method remains near the floor under LOOO; we therefore do not interpret the latter as evidence of successful cross-species organ transfer.

B Spatial-domain recovery

C Full per-cohort results

Table 14 gives the per-cohort breakdown for all seven retained cohorts (HCC/liver, LYMPH_IDC, and READ excluded; see Appendix H) across all six baselines and MIST.

D Cross-organ LOOO transfer, expanded baselines (UNI + CONCH)

Table 15 extends Table 4 with M2OST and EGN under the same UNI+CONCH protocol. EGN attains the highest mean PCC (0.500), followed by MIST at 0.490; their per-cohort ranking varies.

E Cross-organ LOOO transfer with UNI features

Table 16 reports a complementary LOOO comparison using UNI features. EGN has the highest mean PCC (0.493), while MIST reaches 0.477. M2OST narrowly leads EGN on LUNG (0.667 versus 0.666). STEM uses the corrected DiT configuration with $T = 1000$; its SKCM entry contains one completed seed and is marked accordingly.

F Published Per-Cohort Comparison

G Gene Panel Comparison

Table 18 reports intra-cohort mean PCC averaged over seven cohorts and three seeds for three gene-panel strategies: highly variable genes (HVG, the HEST-1k default), unsupervised differentially expressed genes (DEG; Leiden clustering + Wilcoxon), and high-mean highly variable genes (HMHVG; selected using expression mean and variability after variance filtering). The supplied panel-generation script constructs DEG and HMHVG panels once per cohort using all available slides, rather than selecting genes separately within each training fold. These results therefore describe an exploratory comparison with fixed cohort-level panels; held-out expression can influence target selection, and nested check-point validation does not remove this dependence. A strictly inductive comparison requires training-fold-only panel selection and retraining. All methods use identical UNI+CONCH features, nested-validation, and the same training protocol. DEG and HMHVG panels consistently outperform HVG across all methods, with HMHVG yielding the highest mean for every method tested. MIST leads on all three panels, with its advantage most pronounced on HMHVG (0.657 vs. 0.595 for BLEEP) and consistent across DEG (0.624 vs. 0.571) and HVG (0.471 vs. 0.427).

Table 19 gives the per-cohort breakdown. MIST leads on 6 of 7 cohorts under HVG (BLEEP is stronger on COAD) and on all 7 cohorts under both DEG and HMHVG.

H HEST-1k Cohort Statistics

Table 20 lists the number of samples, total spots, and cross-validation folds for each HEST-1k cohort. Two organ groups in HEST-1k contain multiple cohorts: breast (IDC + LYMPH_IDC) and colorectal (COAD + READ). Within each group the constituent

Table 11: Appendix: Intra-cohort gene expression prediction on HEST-1k across gene-panel sizes (Pearson $\uparrow$). PCC@ k denotes prediction with a k -gene panel. Mean_{std} over three seeds with nested-validation; best per cohort and panel in bold. Each panel closes with the cohort-averaged Avg. PCC $\uparrow$, Avg. MSE $\downarrow$, and Avg. MAE $\downarrow$ (best in bold). All methods use identical frozen UNI + CONCH features (1536-dim), splits, and training budget (100 epochs, patience 20, inner-validation selection). The 10-gene panel is the highest-variance subset of each cohort’s HEST 50-HVG panel; the 100-gene panel extends it with the next 50 highest-variance genes. HCC, LYMPH_IDC, and READ are excluded (Appendix H).

Metric	Cohort	ST-Net NBE’20	Hist2ST BIB’22	BLEEP NeurIPS’23	STEM ICLR’25	MIST (Ours)
PCC@10	IDC Breast	0.666 $\pm$ 0.01	0.636 $\pm$ 0.01	0.635 $\pm$ 0.02	0.591 $\pm$ 0.02	0.708$\pm$0.00
	COAD Colorectal	0.255 $\pm$ 0.02	0.447$\pm$0.02	0.413 $\pm$ 0.01	0.266 $\pm$ 0.01	0.382 $\pm$ 0.02
	CCRCC Kidney	0.231 $\pm$ 0.01	0.273 $\pm$ 0.02	0.231 $\pm$ 0.02	0.170 $\pm$ 0.00	0.322$\pm$0.01
	LUNG Lung	0.702 $\pm$ 0.01	0.519 $\pm$ 0.02	0.735 $\pm$ 0.00	0.616 $\pm$ 0.01	0.759$\pm$0.00
	PAAD Pancreas	0.511 $\pm$ 0.04	0.377 $\pm$ 0.07	0.536$\pm$0.03	0.393 $\pm$ 0.04	0.519 $\pm$ 0.05
	PRAD Prostate	0.442 $\pm$ 0.03	0.459 $\pm$ 0.02	0.434 $\pm$ 0.03	0.325 $\pm$ 0.03	0.524$\pm$0.02
	SKCM Skin	0.728 $\pm$ 0.00	0.471 $\pm$ 0.04	0.697 $\pm$ 0.01	0.598 $\pm$ 0.01	0.769$\pm$0.01
	Avg. PCC $\uparrow$	0.505	0.455	0.526	0.423	0.569
	Avg. MSE $\downarrow$	6.119	3.653	3.334	3.647	3.105
	Avg. MAE $\downarrow$	1.897	1.439	1.332	1.397	1.306
PCC@50	IDC Breast	0.551 $\pm$ 0.01	0.309 $\pm$ 0.07	0.519 $\pm$ 0.00	0.439 $\pm$ 0.01	0.581$\pm$0.01
	COAD Colorectal	0.242 $\pm$ 0.01	0.295 $\pm$ 0.01	0.314 $\pm$ 0.01	0.177 $\pm$ 0.01	0.318$\pm$0.01
	CCRCC Kidney	0.163 $\pm$ 0.02	0.184 $\pm$ 0.02	0.175 $\pm$ 0.02	0.123 $\pm$ 0.00	0.252$\pm$0.01
	LUNG Lung	0.518 $\pm$ 0.01	0.470 $\pm$ 0.01	0.562 $\pm$ 0.01	0.423 $\pm$ 0.03	0.589$\pm$0.01
	PAAD Pancreas	0.359 $\pm$ 0.03	0.247 $\pm$ 0.02	0.419 $\pm$ 0.03	0.233 $\pm$ 0.02	0.437$\pm$0.03
	PRAD Prostate	0.294 $\pm$ 0.02	0.333 $\pm$ 0.02	0.348 $\pm$ 0.01	0.207 $\pm$ 0.02	0.410$\pm$0.01
	SKCM Skin	0.665 $\pm$ 0.01	0.522 $\pm$ 0.03	0.645 $\pm$ 0.01	0.491 $\pm$ 0.04	0.707$\pm$0.01
	Avg. PCC $\uparrow$	0.399	0.337	0.426	0.299	0.470
	Avg. MSE $\downarrow$	2.041	1.653	1.445	1.874	1.388
	Avg. MAE $\downarrow$	0.972	0.907	0.843	0.944	0.829
PCC@100	IDC Breast	0.611 $\pm$ 0.01	0.473 $\pm$ 0.07	0.632 $\pm$ 0.01	0.565 $\pm$ 0.01	0.682$\pm$0.01
	COAD Colorectal	0.336 $\pm$ 0.00	0.406 $\pm$ 0.01	0.488 $\pm$ 0.01	0.348 $\pm$ 0.02	0.499$\pm$0.01
	CCRCC Kidney	0.221 $\pm$ 0.01	0.274 $\pm$ 0.03	0.247 $\pm$ 0.02	0.192 $\pm$ 0.00 [†]	0.328$\pm$0.02
	LUNG Lung	0.595 $\pm$ 0.00	0.434 $\pm$ 0.03	0.659 $\pm$ 0.01	0.498 $\pm$ 0.03	0.688$\pm$0.00
	PAAD Pancreas	0.355 $\pm$ 0.04	0.222 $\pm$ 0.04	0.399 $\pm$ 0.04	0.236 $\pm$ 0.02	0.435$\pm$0.03
	PRAD Prostate	0.327 $\pm$ 0.01	0.341 $\pm$ 0.03	0.377 $\pm$ 0.02	0.240 $\pm$ 0.03	0.450$\pm$0.00
	SKCM Skin	0.696 $\pm$ 0.01	0.446 $\pm$ 0.12	0.696 $\pm$ 0.01	0.602 $\pm$ 0.02	0.771$\pm$0.01
	Avg. PCC $\uparrow$	0.449	0.371	0.500	0.415 [†]	0.550
	Avg. MSE $\downarrow$	3.650	1.943	1.690	2.066	1.606
	Avg. MAE $\downarrow$	1.411	1.034	0.948	1.040	0.929

STEM: corrected DiT ($T = 1000$, 34M params). [†]STEM CCRCC at PCC@100 has two of three seeds; its average excludes the pending seed.

Table 12: Cross-organ generalization on mouse HEST (three seeds, mean_{std}). looo: leave-one-organ-out transfer; pooled: five-fold pooled cross-validation. PCC $\uparrow$ on 50 mouse HVGs.

Method	looo PCC $\uparrow$	pooled PCC
ST-Net He et al. (2020)	0.013 _{0.011}	0.357 _{0.036}
Hist2ST Zeng et al. (2022)	0.018 _{0.007}	0.829 _{0.028}
BLEEP Xie et al. (2023)	-0.006 _{0.003}	0.848 _{0.012}
STEM Zhu et al. (2025)	–	0.279 _{0.009}
MIST	0.018_{0.006}	0.887_{0.005}

$\dagger$ All looo values are near the floor because the organ-specific gene panels do not transfer reliably.

Table 13: Spatial-domain recovery. ARI/NMI between domains clustered from predicted and measured expression on held-out slides (mean_{std} over three seeds; higher is better).

Method	ARI $\uparrow$	NMI $\uparrow$
ST-Net He et al. (2020)	0.154 _{0.14}	0.212 _{0.15}
Hist2ST Zeng et al. (2022)	0.134 _{0.11}	0.180 _{0.11}
BLEEP Xie et al. (2023)	0.147 _{0.12}	0.186 _{0.13}
STEM Zhu et al. (2025)	0.109 _{0.12}	0.131 _{0.13}
MIST	0.184_{0.16}	0.239_{0.16}

cohorts differ substantially in size: IDC has $2.25\times$ more spots than LYMPH_IDC (44,974 vs. 19,964), and COAD has $2.2\times$ more spots than READ (18,523 vs. 8,407). To avoid misrepresentation from unweighted averaging over unequal cohorts, Table 11 uses IDC as the representative breast cohort and COAD as the representative colorectal cohort, discarding the smaller partner cohorts (LYMPH_IDC and READ) from the organ-level rows. This choice maximises statistical reliability of each organ-level estimate.

Table 14: Full per-cohort intra-cohort results on HEST-1k (Pearson mean_{std} over three seeds; nested-validation). All methods use identical frozen UNI + CONCH features (1536-dim) and 50-gene panels, except MIST (V3) which uses UNI (1024-dim) for comparison. HCC (liver), LYMPH_IDC, and READ are excluded per Appendix H. Best per row in bold.

Cohort	ST-Net NBE'20	Hist2ST BIB'22	BLEEP NeurIPS'23	M2OST AAAI'25	EGN WACV'23	STEM ICLR'25	MIST (V3)	MIST+CONCH
IDC Breast	0.551±0.01	0.309±0.07	0.519±0.00	0.571±0.01	0.551±0.01	0.439±0.01	0.567±0.00	0.581±0.01
COAD Colorectal	0.242±0.01	0.295±0.01	0.314±0.01	0.328±0.01	0.293±0.01	0.177±0.01	0.299±0.00	0.318±0.01
CCRCC Kidney	0.163±0.02	0.184±0.02	0.175±0.02	0.210±0.02	0.223±0.02	0.123±0.00	0.265±0.02	0.252±0.01
LUNG Lung	0.518±0.01	0.470±0.01	0.562±0.01	0.598±0.00	0.584±0.01	0.423±0.03	0.589±0.01	0.589±0.01
PAAD Pancreas	0.359±0.03	0.247±0.02	0.419±0.03	0.413±0.04	0.415±0.03	0.233±0.02	0.392±0.04	0.437±0.03
PRAD Prostate	0.294±0.02	0.333±0.02	0.348±0.01	0.413±0.01	0.377±0.03	0.207±0.02	0.390±0.02	0.410±0.01
SKCM Skin	0.665±0.01	0.522±0.03	0.645±0.01	0.654±0.01	0.631±0.01	0.491±0.04	0.696±0.00	0.707±0.01
Average	0.399	0.337	0.426	0.455	0.439	0.299	0.457	0.470

MIST (V3) uses UNI (1024-dim); all other methods use UNI + CONCH (1536-dim). STEM: corrected DiT (T=1000, 34M params).

Table 15: Cross-organ LOOO transfer, expanded baselines, per-cohort Pearson $\uparrow$ (HEST-1k) (Pearson mean_{std} over three seeds; nested-validation; liver excluded). Leave-one-organ-out: trained on all other organ groups, tested on the held-out cohort. All methods use identical frozen UNI + CONCH features (1536-dim) and 50-gene panels. Best per row in bold.

Cohort	ST-Net NBE'20	Hist2ST BIB'22	BLEEP NeurIPS'23	M2OST AAAI'25	EGN WACV'23	STEM ICLR'25	MIST+CONCH
IDC Breast	0.198±0.05	0.458±0.01	0.529±0.02	0.437±0.04	0.567±0.02	0.420±0.03	0.520±0.02
COAD Colorectal	0.420±0.06	0.670±0.03	0.662±0.01	0.657±0.05	0.728±0.01	0.452±0.02	0.729±0.03
CCRCC Kidney	0.068±0.01	0.055±0.01	0.095±0.01	0.081±0.03	0.103±0.02	0.090±0.01	0.109±0.01
LUNG Lung	0.493±0.07	0.378±0.18	0.624±0.01	0.665±0.01	0.672±0.01	0.512±0.03	0.675±0.01
PAAD Pancreas	0.459±0.02	0.399±0.02	0.537±0.02	0.513±0.03	0.572±0.01	0.409±0.03	0.548±0.01
PRAD Prostate	0.079±0.00	0.098±0.02	0.118±0.01	0.087±0.02	0.107±0.01	0.046±0.00	0.118±0.01
SKCM Skin	0.520±0.16	0.736±0.01	0.748±0.01	0.740±0.01	0.750±0.02	0.631±0.03	0.728±0.04
Average	0.319	0.399	0.473	0.454	0.500	0.366	0.490

Table 16: Cross-organ LOOO transfer with UNI features (Pearson mean_{std} over three seeds; nested-validation; HCC, LYMPH_IDC, and READ excluded). Leave-one-organ-out: trained on all other organs, tested on the held-out organ. Best per row in bold.

Organ	ST-Net NBE'20	Hist2ST BIB'22	BLEEP NeurIPS'23	M2OST AAAI'25	EGN WACV'23	STEM ICLR'25	MIST (V3)
IDC Breast	0.217±0.06	0.459±0.02	0.509±0.03	0.452±0.01	0.531±0.03	0.406±0.01	0.534±0.02
COAD Colorectal	0.430±0.03	0.688±0.04	0.667±0.02	0.672±0.03	0.749±0.01	0.504±0.03	0.703±0.01
CCRCC Kidney	0.084±0.03	0.084±0.01	0.100±0.02	0.122±0.01	0.106±0.01	0.090±0.01	0.124±0.00
LUNG Lung	0.504±0.04	0.499±0.06	0.621±0.02	0.667±0.01	0.666±0.01	0.561±0.01	0.660±0.01
PAAD Pancreas	0.404±0.05	0.360±0.07	0.521±0.02	0.522±0.01	0.551±0.02	0.460±0.02	0.507±0.04
PRAD Prostate	0.071±0.01	0.104±0.02	0.111±0.00	0.107±0.02	0.098±0.02	0.068±0.01	0.119±0.02
SKCM Skin	0.402±0.08	0.652±0.08	0.745±0.01	0.724±0.02	0.749±0.01	0.686 [†]	0.692±0.06
Average	0.302	0.407	0.468	0.466	0.493	0.396 [†]	0.477

[†]STEM DiT (corrected, T=1000, 100 epochs) LOOO: SKCM (skin) has only 1 seed complete; all other organs use 3 seeds. Average marked [†] accordingly.

Table 17: Published per-cohort accuracy on the standard HEST-1k splits, mean PCC $\uparrow$ as reported in respective papers (ten cohorts, no organ grouping). Shown for context only (not comparable to Table 11), which uses organ grouping and nested-validation. [†]MIST uses our organ-grouped splits, not the standard per-cohort split.

Method	HEST-1k (10 cohorts)
ST-Net He et al. (2020) (NBE'20)	0.286
Hist2ST Zeng et al. (2022) (BIB'22)	–
MCToGene Sun et al. (2026) (CVPR'26)	0.435
MIST [†] (ours)	0.440

Table 18: Gene panel comparison: intra-cohort mean $PCC\uparrow$ (HEST-1k). $Mean_{std}$ over three seeds; per-seed value is the average over seven cohorts. All methods use identical frozen UNI + CONCH features (1536-dim) and nested-validation. Best per column in bold.

Method	HVG	DEG	HMHVG
ST-Net He et al. (2020)	0.398 _{0.004}	0.523 _{0.004}	0.550 _{0.008}
Hist2ST Zeng et al. (2022)	0.330 _{0.032}	0.494 _{0.014}	0.544 _{0.038}
BLEEP Xie et al. (2023)	0.427 _{0.004}	0.571 _{0.005}	0.595 _{0.014}
STEM Zhu et al. (2025)	0.291 _{0.005}	0.456 _{0.002}	0.485 _{0.010}
MIST (ours)	0.471_{0.005}	0.624_{0.002}	0.657_{0.006}

Table 19: Per-cohort gene panel comparison: intra-cohort $PCC\uparrow$ (HEST-1k). $Mean_{std}$ over three seeds. All methods use UNI + CONCH (1536-dim) and nested-validation. Best per cohort per panel in bold.

Panel	Cohort	ST-Net	Hist2ST	BLEEP	STEM	MIST
HVG	IDC Breast	0.551 _{0.007}	0.305 _{0.108}	0.519 _{0.002}	0.434 _{0.002}	0.581_{0.008}
	COAD Colorectal	0.235 _{0.005}	0.298 _{0.009}	0.322_{0.006}	0.177 _{0.014}	0.318 _{0.008}
	CCRCC Kidney	0.163 _{0.015}	0.180 _{0.030}	0.175 _{0.015}	0.118 _{0.006}	0.252_{0.007}
	LUNG Lung	0.521 _{0.010}	0.430 _{0.050}	0.565 _{0.008}	0.407 _{0.003}	0.589_{0.006}
	PAAD Pancreas	0.366 _{0.032}	0.248 _{0.038}	0.412 _{0.034}	0.233 _{0.019}	0.437_{0.028}
	PRAD Prostate	0.287 _{0.023}	0.323 _{0.012}	0.346 _{0.007}	0.207 _{0.016}	0.410_{0.006}
	SKCM Skin	0.661 _{0.006}	0.523 _{0.003}	0.650 _{0.004}	0.461 _{0.018}	0.707_{0.007}
	Average	0.398	0.330	0.427	0.291	0.471
DEG	IDC Breast	0.697 _{0.010}	0.549 _{0.067}	0.731 _{0.014}	0.684 _{0.020}	0.774_{0.009}
	COAD Colorectal	0.497 _{0.003}	0.608 _{0.010}	0.639 _{0.016}	0.534 _{0.014}	0.677_{0.003}
	CCRCC Kidney	0.312 _{0.014}	0.335 _{0.013}	0.304 _{0.003}	0.253 _{0.009}	0.402_{0.013}
	LUNG Lung	0.566 _{0.006}	0.518 _{0.006}	0.621 _{0.006}	0.440 _{0.009}	0.662_{0.002}
	PAAD Pancreas	0.425 _{0.032}	0.352 _{0.011}	0.487 _{0.029}	0.306 _{0.042}	0.518_{0.026}
	PRAD Prostate	0.413 _{0.004}	0.437 _{0.008}	0.466 _{0.005}	0.355 _{0.026}	0.539_{0.012}
	SKCM Skin	0.752 _{0.009}	0.656 _{0.004}	0.746 _{0.026}	0.617 _{0.023}	0.794_{0.002}
	Average	0.523	0.494	0.571	0.456	0.624
HMHVG	IDC Breast	0.724 _{0.014}	0.633 _{0.082}	0.739 _{0.018}	0.687 _{0.008}	0.774_{0.011}
	COAD Colorectal	0.503 _{0.010}	0.659 _{0.018}	0.645 _{0.004}	0.528 _{0.032}	0.698_{0.005}
	CCRCC Kidney	0.339 _{0.007}	0.387 _{0.010}	0.349 _{0.010}	0.290 _{0.013}	0.454_{0.017}
	LUNG Lung	0.568 _{0.005}	0.598 _{0.003}	0.661 _{0.002}	0.517 _{0.011}	0.693_{0.004}
	PAAD Pancreas	0.441 _{0.032}	0.341 _{0.038}	0.493 _{0.023}	0.305 _{0.027}	0.524_{0.023}
	PRAD Prostate	0.452 _{0.002}	0.477 _{0.011}	0.432 _{0.045}	0.337 _{0.016}	0.555_{0.022}
	SKCM Skin	0.824 _{0.029}	0.710 _{0.141}	0.849 _{0.018}	0.731 _{0.060}	0.901_{0.002}
	Average	0.550	0.544	0.595	0.485	0.657

Table 20: HEST-1k cohort statistics. Spots counted after barcode filtering. Folds = number of train/test splits in the per-cohort cross-validation. HCC (liver) is excluded from Table 11 due to its near-zero signal floor.

Cohort	Organ	Samples	Spots	Folds
IDC	Breast	4	44,974	4
LYMPH_IDC	Breast	4	19,964	4
COAD	Colorectal	4	18,523	2
READ	Colorectal	4	8,407	2
CCRCC	Kidney	24	74,220	6
HCC	Liver*	2	4,248	2
LUNG	Lung	2	7,505	2
PAAD	Pancreas	3	9,793	3
PRAD	Prostate	23	62,710	2
SKCM	Skin	2	5,716	2

*HCC (liver) is included here for completeness but excluded from Table 11; all baselines score near zero on this cohort due to low spatial gene-expression variance.

I Proofs of the formal guarantees

I.1 Proof of Proposition 1

For every pair of spots i, j , orthogonality of R gives

$$\|R\mathbf{p}_i + \mathbf{t} - (R\mathbf{p}_j + \mathbf{t})\|_2 = \|R(\mathbf{p}_i - \mathbf{p}_j)\|_2 \quad (9)$$

$$= \|\mathbf{p}_i - \mathbf{p}_j\|_2. \quad (10)$$

Hence, transforming the coordinate frame leaves the complete distance matrix unchanged. It also leaves the k NN graph, radial basis features, and distance-dependent attention biases unchanged. The input projection is unchanged because $\mathbf{F}$ is fixed. At every layer, the local pathway therefore receives the same features, neighbors, and distance biases, while the global and slide pathways do not use coordinates. Their outputs, the residual fusion, and the feed-forward update are identical. Induction over the L layers gives identical final representations and predictions, proving Proposition 1.

I.2 Proof of Proposition 2

Let $\mathbf{F}' = Q\mathbf{F}$ and $\mathbf{P}' = Q\mathbf{P}$. The pairwise distance matrix is relabelled as $\mathbf{D}' = Q\mathbf{D}Q^\top$. Under equivariant tie handling, the k NN graph and its radial basis edge features are relabelled by the same permutation. The input projection and all pointwise linear maps, layer normalizations, activations, and feed-forward networks commute with Q .

For local attention, permuting the spots only relabels each neighborhood and its attention sum, so the local output is multiplied by Q . Global self-attention is likewise permutation equivariant because its query-key score matrix is transformed by Q on both axes and its value matrix by Q . The slide attention scores are permuted, but their weighted sum is unchanged; broadcasting that same summary to all spots is equivariant. Consequently, residual fusion preserves equivariance at every layer. Induction over the layers and application of the pointwise prediction head yield $f_\theta(Q\mathbf{F}, Q\mathbf{P}) = Qf_\theta(\mathbf{F}, \mathbf{P})$, proving Proposition 2.

J Three-Metric Intra-Cohort Summary

Table 21 extends the intra-cohort comparison with Spearman correlation (SCC) and mean squared error (MSE) alongside PCC, averaged over seven organ groups.

Table 21: Intra-cohort three-metric summary (HEST-1k, averaged over seven organ groups; three seeds; UNI + CONCH features). PCC $\uparrow$, SCC $\uparrow$, MSE $\downarrow$. Best per column in bold.

Method	PCC $\uparrow$	SCC $\uparrow$	MSE $\downarrow$
ST-Net He et al. (2020) (NBE'20)	0.399	0.356	2.043
Hist2ST Zeng et al. (2022) (BIB'22)	0.337	0.321	1.616
BLEEP Xie et al. (2023) (NeurIPS'23)	0.426	0.388	1.464
STEM Zhu et al. (2025) (ICLR'25)	0.299	0.274	1.855
MIST (ours)	0.470	0.430	1.389

K Per-Cohort MSE and MAE (Intra-Cohort)

Tables 22 and 23 report the per-cohort mean squared error (MSE) and mean absolute error (MAE) for the same models, cohorts, and protocol as the intra-cohort Pearson comparison in Table 11. MIST attains the best average on both error metrics (MSE 1.389, MAE 0.829), consistent with its Pearson result; per-cohort leadership is less uniform than on Pearson, as the training objective couples MSE with a correlation term.

Table 22: Intra-cohort MSE $\downarrow$ on HEST-1k, per cohort. Mean_{std} over three seeds with nested-validation; best (lowest) per row in bold. Same models, cohorts, features (frozen UNI + CONCH, 1536-dim), 50-gene panels, and training budget as Table 11.

Cohort	ST-Net NBE'20	Hist2ST BIB'22	BLEEP NeurIPS'23	STEM ICLR'25	MIST (Ours)
IDC Breast	2.966 $\pm$ 0.311	2.740 $\pm$ 0.194	2.503 $\pm$ 0.061	2.811 $\pm$ 0.056	2.242$\pm$0.031
COAD Colorectal	2.335 $\pm$ 0.063	2.195$\pm$0.058	2.201 $\pm$ 0.062	2.632 $\pm$ 0.021	2.322 $\pm$ 0.031
CCRCC Kidney	0.587 $\pm$ 0.007	0.578 $\pm$ 0.032	0.604 $\pm$ 0.011	0.763 $\pm$ 0.016	0.556$\pm$0.019
LUNG Lung	3.388 $\pm$ 0.062	1.732 $\pm$ 0.003	1.650$\pm$0.024	2.036 $\pm$ 0.052	1.665 $\pm$ 0.043
PAAD Pancreas	2.051 $\pm$ 0.235	1.331 $\pm$ 0.193	1.151$\pm$0.258	1.711 $\pm$ 0.206	1.173 $\pm$ 0.166
PRAD Prostate	0.586 $\pm$ 0.011	0.560 $\pm$ 0.020	0.570 $\pm$ 0.019	0.822 $\pm$ 0.014	0.537$\pm$0.034
SKCM Skin	2.387 $\pm$ 0.085	2.176 $\pm$ 0.016	1.570 $\pm$ 0.036	2.207 $\pm$ 0.260	1.227$\pm$0.030
Average	2.043	1.616	1.464	1.855	1.389

Table 23: Intra-cohort MAE $\downarrow$ on HEST-1k, per cohort. Mean_{std} over three seeds with nested-validation; best (lowest) per row in bold. Same models, cohorts, features (frozen UNI + CONCH, 1536-dim), 50-gene panels, and training budget as Table 11.

Cohort	ST-Net NBE'20	Hist2ST BIB'22	BLEEP NeurIPS'23	STEM ICLR'25	MIST+CONCH
IDC Breast	1.273 $\pm$ 0.066	1.246 $\pm$ 0.032	1.141 $\pm$ 0.022	1.210 $\pm$ 0.015	1.096$\pm$0.010
COAD Colorectal	0.996 $\pm$ 0.014	1.008 $\pm$ 0.017	0.986$\pm$0.020	1.088 $\pm$ 0.008	1.040 $\pm$ 0.012
CCRCC Kidney	0.575 $\pm$ 0.003	0.574 $\pm$ 0.018	0.583 $\pm$ 0.003	0.612 $\pm$ 0.011	0.569$\pm$0.008
LUNG Lung	1.373 $\pm$ 0.014	0.985 $\pm$ 0.002	0.960 $\pm$ 0.008	1.050 $\pm$ 0.003	0.957$\pm$0.014
PAAD Pancreas	1.013 $\pm$ 0.051	0.872 $\pm$ 0.055	0.796$\pm$0.088	0.936 $\pm$ 0.054	0.798 $\pm$ 0.046
PRAD Prostate	0.564 $\pm$ 0.004	0.546$\pm$0.003	0.547 $\pm$ 0.007	0.620 $\pm$ 0.014	0.550 $\pm$ 0.017
SKCM Skin	1.015 $\pm$ 0.023	1.056 $\pm$ 0.015	0.926 $\pm$ 0.011	1.049 $\pm$ 0.070	0.791$\pm$0.005
Average	0.973	0.898	0.848	0.938	0.829

L Cross-Organ Generalization: PCC and MSE

Table 24 reports both PCC and MSE for the LOOO and POOLED cross-organ regimes, complementing the PCC-only summaries in the main paper.

Table 24: Cross-organ generalization: PCC and MSE (three seeds, mean_{std}). looo: leave-one-organ-out transfer; pooled: 5-fold pooled cross-validation. PCC $\uparrow$ / MSE $\downarrow$ on HEST-1k.

Method	looo		pooled	
	PCC $\uparrow$	MSE $\downarrow$	PCC $\uparrow$	MSE $\downarrow$
ST-Net He et al. (2020) (NBE'20)	0.267 _{0.02}	1.846 _{0.10}	0.370 _{0.01}	0.907 _{0.05}
Hist2ST Zeng et al. (2022) (BIB'22)	0.360 _{0.02}	1.182 _{0.07}	0.737 _{0.01}	0.488 _{0.01}
BLEEP Xie et al. (2023) (NeurIPS'23)	0.416 _{0.01}	1.010_{0.02}	0.758 _{0.01}	0.427_{0.02}
STEM Zhu et al. (2025) (ICLR'25)	0.322 _{0.01}	1.274 _{0.00}	0.643 _{0.01}	0.594 _{0.01}
MIST (ours)	0.421_{0.01}	1.233 _{0.11}	0.760_{0.01}	0.440 _{0.02}

M Intra-Cohort Results with UNI + CONCH Features, Extended Baselines

Table 25 extends the intra-cohort comparison to include M2OST and EGN under the same UNI + CONCH feature protocol.

Table 25: Intra-cohort results using UNI+CONCH features, extended baselines (Pearson mean_{std} over three seeds; nested-validation). All methods use identical frozen UNI+CONCH features (1536-dim) and 50-gene panels. HCC (liver), LYMPH_IDC, and READ are excluded per Appendix H. Best per row in bold.

Cohort	ST-Net	NBE'20	Hist2ST	BIB'22	BLEEP	NeurIPS'23	M2OST	AAAI'25	EGN	WACV'23	STEM	ICLR'25	MIST (ours)
IDC Breast	0.551±0.01	0.309±0.06	0.519±0.00	0.571±0.01	0.551±0.01	0.439±0.01	0.581±0.01						
COAD Colorectal	0.242±0.01	0.295±0.01	0.314±0.01	0.328±0.01	0.293±0.01	0.177±0.01	0.318±0.01						
CCRCC Kidney	0.163±0.02	0.184±0.02	0.175±0.02	0.210±0.02	0.223±0.02	0.123±0.00	0.252±0.01						
LUNG Lung	0.518±0.01	0.470±0.01	0.562±0.01	0.598±0.00	0.584±0.01	0.423±0.03	0.589±0.01						
PAAD Pancreas	0.359±0.03	0.247±0.02	0.419±0.03	0.413±0.04	0.415±0.03	0.233±0.02	0.437±0.03						
PRAD Prostate	0.294±0.02	0.333±0.02	0.348±0.01	0.413±0.01	0.377±0.03	0.207±0.02	0.410±0.01						
SKCM Skin	0.665±0.01	0.522±0.03	0.645±0.01	0.654±0.01	0.631±0.01	0.491±0.04	0.707±0.01						
Average	0.399	0.337	0.426	0.455	0.439	0.299	0.470						

All entries use three completed seeds. STEM uses the corrected DiT configuration ($T = 1000$, 100 epochs).

N Training Objective Ablation

Table 26 isolates the contribution of the Pearson correlation term in the training objective from the architectural design, confirming that the accuracy gap over baselines is not attributable to the loss function alone.

Table 26: Objective ablation (HEST-1k). Effect of adding the Pearson correlation term to the MSE objective; averaged over seven organ groups, three seeds.

Model	Objective	PCC ↑
MIST	MSE only	0.382
MIST	MSE + PCC	0.389